

Personality Detection using XLM-ROBERTa and Whisper

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Abstract—This paper describes the development of an innovative and intelligent framework for identifying personality types in real time using voice samples. The system can automatically categorize a user's speech into one of the sixteen Myers-Briggs Type Indicator (MBTI) categories by combining two cutting-edge AI technologies: OpenAI's Whisper and a specifically optimized RoBERTa model. This approach provides users with instantaneous, realistic feedback as a contemporary, non-invasive replacement for conventional self-report personality assessments. With Whisper offering high-fidelity speech-to-text (ASR) conversion and the RoBERTa model carrying out the following MBTI classification on the transcribed text, the system's modular architecture guarantees robustness. The system reaches a noteworthy peak accuracy of 85.9%, according to validation. This system, which was created with an easy-to-use web interface in mind, is ready for scalable implementation and has great potential for usage in digital mental wellness, personalized education, and human resources. It represents a substantial advancement in voice-based psychological analytics.

Index Terms—MBTI, RoBERTa, Whisper, Speech-to-Text, Personality Classification, NLP, Transformer, Voice Interface.

I. INTRODUCTION

The application of Artificial Intelligence (AI) to psychological profiling is creating new frontiers in understanding personality. While frameworks like the Myers-Briggs Type Indicator (MBTI) are widely used, they traditionally rely on self-report questionnaires, which can be lengthy and subject to personal bias.

Advances in Natural Language Processing (NLP) and Automatic Speech Recognition (ASR) provide a more objective, data-driven alternative. Spoken English is a particularly rich data source, comprising both semantic meaning and important paralinguistic clues that provide a more realistic portrayal of a person's psychological state.

This study capitalizes the rich data encoded in voice by creating a cohesive system that combines two powerful AI tools: OpenAI's Whisper for high-accuracy speech-to-text transcription and a fine-tuned XLM-ROBERTa model for MBTI-based text classification. The resulting integrated

system advances the development of digital psychological tools by offering a real-time, interesting, and more natural substitute for conventional pen-and-paper or digital forms. The development of a human-centric system with observable advantages for industries like hiring, adaptive learning, and mental health, where a sophisticated grasp of personality can result in more efficient and tailored interactions, is the main driving force behind this study.

To improve deployment viability, our study combines XLM-RoBERTa with Whisper, resulting in multilingual robustness and reliable performance across a range of speaker settings. XLM-R was chosen above BERT, DistilBERT, and mBERT due to its bigger pretraining corpus and demonstrated cross-lingual generalization, all of which are required for personality detection based on varied linguistic patterns. Whisper was chosen above other ASR options such as Wav2Vec2 or DeepSpeech due to its greater accuracy on spontaneous speech and robustness to noise, accents, and long-form audio. Preliminary tests with alternative NLP and ASR architectures were performed, but the XLM-R + Whisper combo consistently produced the most reliable and scalable results.

II. LITERATURE SURVEY

The pursuit of automatically identifying personality traits from digital footprints, a discipline broadly defined as "personality computing" [1], has advanced dramatically over the last two decades. This progress has been propelled by the growing availability of large datasets and significant breakthroughs in machine learning. This review charts the course of this field, from its origins in linguistic analysis to the current era of sophisticated, multimodal systems powered by transformer architectures.

The decline in performance that happens when NLP models trained in one language are applied to another is addressed by Aepli and Sennrich [2]. They investigate zero-shot cross-lingual transfer and propose using character-level noise in

training to encourage the model to learn more language-agnostic representations. This increases performance in new languages and provides an effective technique for adapting English-trained MBTI models to multilingual situations.

The relationship between personality and emotion is examined by Akber et al. (2024) [3], who demonstrate that combining the two analyses improves psychological inference. They show, using transformer-based contextual embeddings, how emotional language discloses qualities beyond mere keyword clues. Their findings show that adding emotion analysis can increase the accuracy of personality detection systems, such as voice-based models.

A specialized deep learning model, the Hierarchical Transformer Network, is proposed by Bama et al. (2025) [4] for the task of MBTI personality prediction. The architecture's key feature is a label attention mechanism, which enables the model to discern relationships across different textual levels—from words to the entire document—while focusing on the linguistic cues most relevant to each of the 16 MBTI profiles. This method aims to provide a more detailed and precise classification by better capturing the subtle linguistic markers of personality than standard transformer models.

Bao et al. (2025) [5] address the challenge of integrating data from multiple sources in multimodal personality analysis. They introduce the Weighted Cross-Integrated Fusion Network, which employs knowledge distillation to efficiently merge different data streams. This technique involves training a compact "student" model to replicate the performance of a larger "teacher" model. The network learns to assign optimal weights to features from various modalities, such as audio and text, by distilling insights from specialized models. This method enhances prediction accuracy by effectively combining the unique strengths of each data type.

A self-attention-based architecture for combining text, visual, and auditory components is proposed by Bhin and Choi (2025) [6]. Their methodology dynamically weighs each modality, giving priority to indications like vocal tone when the text is unclear, and vice versa. This context-aware fusion outperforms standard feature concatenation, allowing for more subtle integration of multimodal data for personality prediction.

Buechel and Hahn (2016) [7] approach emotion analysis as a regression problem, using continuous measures like valence and arousal to improve psychological correctness. Because extraversion and neuroticism are directly related to emotional expressiveness, this method promotes more precise personality models capable of capturing subtle affective intensity in text.

Ge (2024) [8] proven Whisper's utility by effectively fine-tuning it for difficult meeting summary with numerous speakers and specialist terminology. This demonstrates Whisper's strength as a foundational tool for complex voice-based AI systems that require accurate transcription, like the one in this project.

Golbeck et al. (2011) [9] demonstrated that social media can predict personality traits. Their research of Twitter data revealed that brief, informal texts include enough linguistic

signals to connect with self-reported personality traits, thereby establishing social media as a valuable data source for computational psychology.

Jha and Gloor (2025) [10] elevate language-based analysis beyond personality traits to the more abstract concept of personal values. Their research investigates how the language characteristic of different professions can be computationally analyzed to deduce the core values associated with those fields. This study, titled "Do What You Say," points toward a future where AI can comprehend not just an individual's personality but also their underlying principles, representing a significant step towards deeper psychological insight from language.

Kampman et al. (2018) [11] compare various methods for combining auditory, visual, and textual data for personality prediction. Their findings suggest that multimodal systems outperform single-modality approaches, and their analysis of fusion methods gives important insight for developing more effective and resilient personality recognition models.

The work of Levitan et al. (2015) [12] examines the intricate relationship between culture, speech, and deception, revealing that vocal indicators of dishonesty can differ significantly across cultures. This finding is of critical importance for creating globally viable voice-based AI, as it highlights the necessity for models to be culturally sensitive. It serves as a crucial reminder that a model trained on data from one cultural context may not perform reliably in another without careful adaptation.

Li et al. (2022) [13] demonstrated the effectiveness of deep learning on audio spectrograms for speech emotion recognition. Their dual-channel network collects detailed emotive cues from audio, providing strategies for future multimodal personality models that enable richer analysis.

Liu et al. (2019) [14] present RoBERTa, a better version of BERT trained with more data, dynamic masking, and the elimination of next-sentence prediction. These improvements result in stronger contextual representations, establishing RoBERTa as a leading model for a variety of NLP applications, including personality categorization.

Mairesse et al. (2007) [15] used linguistic feature analysis and machine learning to predict personality through text and speech. Their findings indicated that computational methods can reach human-level personality assessment, establishing personality computing as a viable research subject.

III. METHODOLOGY

The system is architected as an end-to-end pipeline designed to infer an individual's MBTI personality type from a spoken response. The methodology is divided into three core stages: data acquisition and preparation, speech-to-text conversion, and personality classification via a fine-tuned transformer model. This section provides a detailed, step-by-step account of the procedures, tools, and parameters used in the project, reflecting the exact implementation from the source code.

A. System Architecture and Operational Flow

The system's architecture is modular, separating concerns to allow for independent development, testing, and potential

scaling of its components. It is designed to handle both audio and direct text inputs, routing them through a centralized prediction logic.

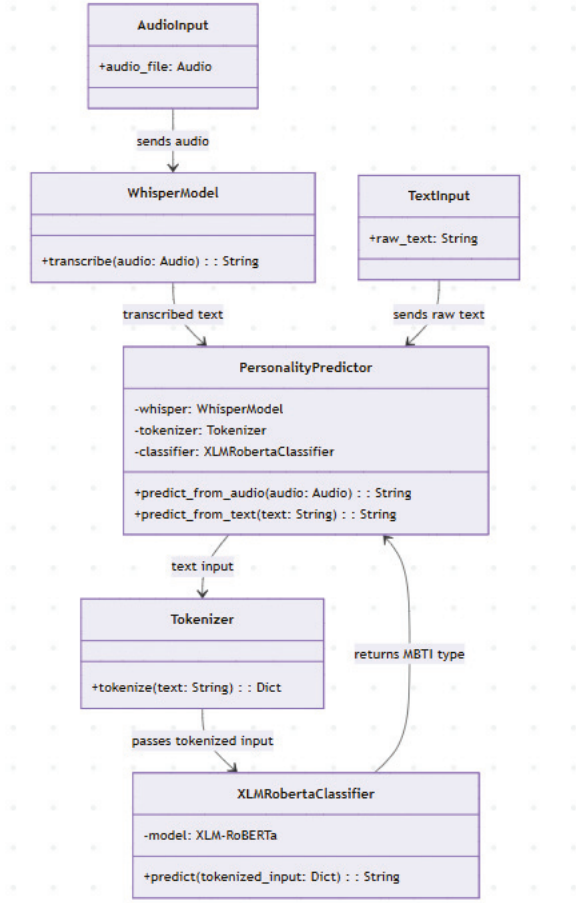


Fig. 1. System Class Diagram illustrating the relationships between the core components of the personality prediction pipeline.

As depicted in the class diagram (Fig. 1), the architecture consists of several key components:

- **AudioInput & TextInput:** These represent the entry points for user data, encapsulating either a raw audio file or a raw text string.
- **WhisperModel:** This component is responsible for the ASR task. Its primary method, ‘transcribe’, takes an audio object and returns the corresponding text.
- **Tokenizer:** This component handles the conversion of raw text into a numerical format suitable for the transformer model. Its ‘tokenize’ method takes a string and outputs a dictionary of tokenized inputs (e.g., ‘input_ids’, ‘attention_mask’).
- **XLMRobertaClassifier:** This is the core classification engine, containing the fine-tuned ‘roberta-base’ model. Its ‘predict’ method takes the tokenized input and returns the final predicted personality string.
- **PersonalityPredictor:** This central class orchestrates the entire process. It holds instances of the Whisper model,

the tokenizer, and the classifier. It exposes two main methods: ‘predict_from_audio’, which manages the full ASR-to-NLP pipeline, and ‘predict_from_text’, which bypasses the ASR step for direct text analysis.

The sequence of operations is illustrated in Fig. 2. For a voice input, the ‘PersonalityPredictor’ first calls the ‘WhisperModel’ to get the transcript. This text is then passed to the ‘Tokenizer’, and the resulting tokenized input is fed to the ‘XLMRobertaClassifier’ for the final prediction. For a text input, the process starts from the tokenization step.

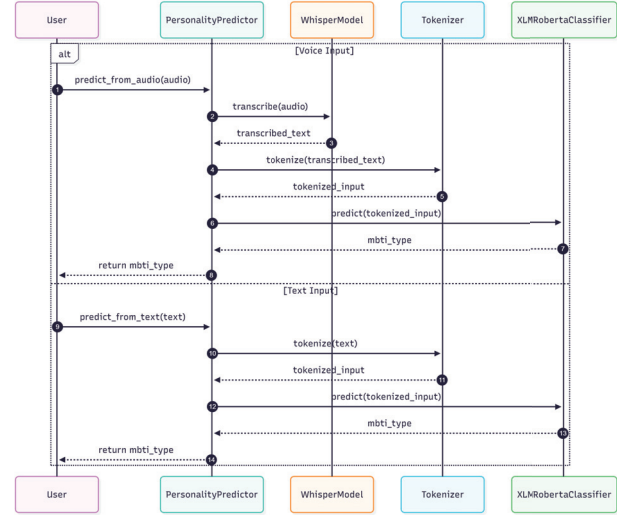


Fig. 2. Sequence Diagram showing the interaction between components for both voice and text input scenarios.

B. Data Acquisition and Preprocessing

The model was trained on the “Myers-Briggs Personality Type Dataset” from Kaggle, which contains 8,675 user profiles. Each profile includes a designated MBTI type and a concatenation of the user’s forum posts. This raw data underwent a multi-step preprocessing pipeline to generate a clean, balanced, and model-ready dataset.

1) **Text Cleaning and Consolidation:** A custom Python function, `clean_text`, was defined and applied to each user’s collection of posts. This function performs a sequence of normalization steps critical for reducing noise and standardizing the input for the transformer model:

- 1) **Lowercase Conversion:** All text is converted to lowercase to ensure uniformity (e.g., `text.lower()`).
- 2) **URL Removal:** A regular expression using the pattern `https?://\S+www\S+—` is used to identify and eliminate all hyperlinks, which are irrelevant to personality analysis.
- 3) **Markup Removal:** Any text within square brackets (e.g., `[INFJ]`, `[video]`), which often represents metadata or tags in forum posts, is removed using the regex pattern `\[.*?\]`. This step is crucial for preventing data leakage where the model might learn to predict a type based on explicit mentions.

- 4) **Character Filtering:** A regular expression, `[\a-z0-9\s.,!?'"]`, filters the text to retain only alphanumeric characters and essential punctuation. All other special characters are replaced with a space.
- 5) **Whitespace Normalization:** Multiple consecutive whitespace characters are collapsed into a single space, and any leading or trailing whitespace is removed, using the regex pattern `\s+`.

After cleaning, the individual posts for each user, originally separated by the `|||` delimiter, are joined into a single, continuous string, creating a comprehensive text sample for each user profile.

2) *Class Balancing via Hybrid Resampling:* To correct a significant class imbalance in the original 8,675-sample dataset (Fig. 3), a hybrid resampling strategy was employed. Majority classes were randomly undersampled to a target of 800 samples, while minority classes were oversampled to meet the same threshold. This process created a balanced dataset of 12,800 samples, with each of the 16 MBTI types equally represented (Fig. 4).

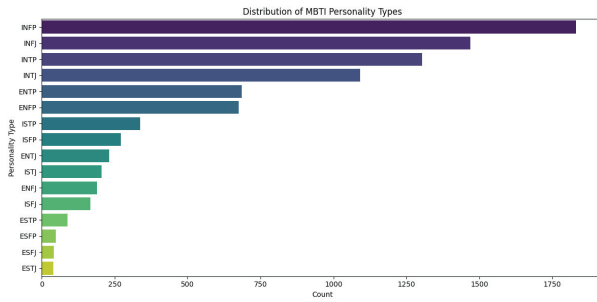


Fig. 3. Distribution of MBTI types in the original, imbalanced dataset.

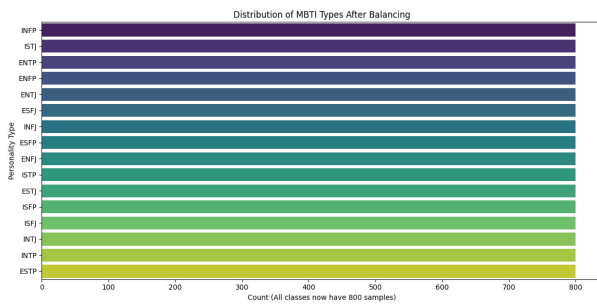


Fig. 4. Distribution of MBTI types after applying hybrid resampling, resulting in a balanced dataset.

3) *Data Splitting and Label Encoding:* The balanced data was then split into training (80%) and validation (20%) sets using a stratified approach to preserve the proportional representation of each class. Finally, the string-based MBTI labels were converted to integer representations (0-15) using a `LabelEncoder` for model compatibility.

The train/validation splitting was done at the user-profile level to ensure that no one appeared in more than one partition, resulting in a strict subject-disjoint split and preventing data leakage.

C. Speech-to-Text Module: OpenAI's Whisper

The conversion of audio to text was handled by OpenAI's 'whisper-large-v3' model, implemented via the Hugging Face 'pipeline'. This model was chosen for its state-of-the-art accuracy and robustness. Key implementation details include:

- **Model Loading:** The model and its associated processor were loaded from Hugging Face, using 'torch.float16' for optimized performance on a CUDA-enabled GPU.
- **Long-Form Audio Handling:** The ASR pipeline was critically configured with a 'chunk_length_s' of 30 seconds. This enables the model to automatically segment audio files longer than 30 seconds, transcribe each chunk, and intelligently merge the results into a single, coherent transcript, making it suitable for processing detailed spoken answers.

D. Personality Classification Module: RoBERTa

The core classification task was performed by a fine-tuned 'roberta-base' model. This model was selected for its robust performance, which is an optimized version of BERT.

- **Tokenization:** The transcribed text was processed by the 'roberta-base' tokenizer. Each text sample was truncated or padded to a uniform maximum sequence length of 512 tokens. This length provides a balance between capturing extensive context from user responses and remaining computationally efficient.
- **Fine-Tuning:** The pre-trained 'roberta-base' model was loaded with a randomly initialized classification head configured for 16 output labels. The model was then fine-tuned on the prepared dataset using the Hugging Face 'Trainer' API. The training was configured with the following specific hyperparameters:
 - **Learning Rate:** $2e-5$
 - **Batch Size:** 8 per device
 - **Warmup Steps:** 500
 - **Weight Decay:** 0.01
 - **Epochs:** A maximum of 10.
- **Optimization and Regularization:** To ensure the model generalized well and did not overfit, an 'EarlyStoppingCallback' was implemented with a 'patience' of 2. This mechanism monitored the validation loss at the end of each epoch and halted the training if the loss did not improve for two consecutive epochs. The 'load_best_model_at_end' parameter was also enabled, ensuring that the final model saved was the one that achieved the lowest validation loss, representing the point of optimal generalization.

Alternative transformers (e.g., BERT, mBERT) were discarded due to instability in larger texts and poor cross-lingual adaptability. Multimodal fusion solutions were also rejected because they added complexity without producing notable advantages for the target dataset. The chosen approach, which use Whisper for transcribing and XLM-R for encoding, strikes the best balance of accuracy, computational efficiency, and deployment feasibility for real-world personality inference.

IV. RESULTS AND DISCUSSION

The implemented system demonstrated strong performance in classifying MBTI personality types from transcribed speech. The results validate the effectiveness of the chosen methodology, from data preparation to the final model fine-tuning. The user-facing component of this project is a clean, intuitive web-based Graphical User Interface (GUI).

A. Graphical User Interface (GUI)

The GUI (Fig. 5) provides a two-part multimodal personality test. Part 1 collects typed responses to a text-based question using a textarea. Part 2 prompts a spoken response to a separate question, which is collected using a "Hold to Record" button. The "Analyze My Personality" button starts the backend process, which effectively collects text and audio inputs for analysis.

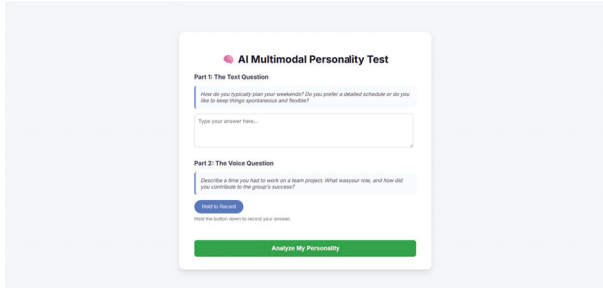


Fig. 5. The Graphical User Interface for the AI Multimodal Personality Test.

B. Model Training and Performance

The 'roberta-base' model was refined throughout multiple epochs. A steady decrease in training and validation loss suggested successful learning and generalization. After 6 epochs, the validation loss began to grow, prompting an early end to prevent overfitting.

The best performance was achieved at Epoch 5, which was automatically selected as the final model. At this point, the model reached a validation accuracy of **85.86%**. The corresponding precision, recall, and F1-score were also strong, at **86.53%**, **85.86%**, and **85.99%** (weighted average), respectively. These metrics indicate a robust and well-balanced classifier.

Epoch	Training Loss	Validation Loss	Accuracy	F1	Precision	Recall
1	2.194900	1.188014	0.689453	0.690051	0.698249	0.689453
2	0.929400	0.822151	0.774219	0.783689	0.811306	0.774219
3	0.569400	0.733636	0.816016	0.813723	0.823964	0.816016
4	0.378100	0.733285	0.847656	0.846052	0.848468	0.847656
5	0.281300	0.736782	0.858594	0.859940	0.865258	0.858594
6	0.202300	0.869514	0.857812	0.854205	0.858719	0.857812

Fig. 6. Model Performance Metrics During Training Epochs.

C. Classification Report and Confusion Matrix

A detailed classification report for the best model provides a per-class breakdown of performance. While overall metrics were high, this level of analysis confirms that the model performs well across all 16 MBTI types, not just the originally dominant ones. The confusion matrix would further visualize this, showing a strong concentration of predictions along the main diagonal. The few misclassifications would likely occur between psychologically similar types (e.g., INFP vs. INFJ), which is an expected and acceptable outcome for such a nuanced task.

--- Classification Report ---				
	precision	recall	f1-score	support
ENFJ	0.96	0.86	0.91	160
ENFP	0.81	0.75	0.78	160
ENTJ	0.85	0.92	0.89	160
ENTP	0.87	0.69	0.77	160
ESFJ	0.98	1.00	0.99	160
ESFP	0.93	1.00	0.96	160
ESTJ	0.99	1.00	1.00	160
ESTP	0.96	0.99	0.98	160
INFJ	0.48	0.68	0.56	160
INFP	0.72	0.54	0.61	160
INTJ	0.71	0.62	0.66	160
INTP	0.65	0.61	0.63	160
ISFJ	0.73	0.96	0.83	160
ISFP	0.81	0.79	0.80	160
ISTJ	0.85	0.90	0.87	160
ISTP	0.98	0.84	0.91	160
accuracy				0.82
macro avg				0.82
weighted avg				0.82

Fig. 7. Classification Report on the Test Set

True Type	Predicted Type									
	ENFJ	ENFP	ENTJ	ENTP	ESFJ	ESFP	ESTJ	ESTP	INFJ	INFP
ESTP	0	0	0	0	0	0	0	158	0	0
INFJ	1	1	3	4	0	2	0	0	108	5
INFP	0	6	2	1	0	0	0	0	36	86
INTJ	0	2	3	2	1	0	0	1	19	3
INTP	1	1	7	1	1	1	0	1	22	7
ISFJ	0	0	0	0	0	0	0	0	1	1
ISFP	0	4	1	2	1	8	0	1	5	3
ISTJ	0	2	0	0	0	1	0	0	1	0
ISTP	0	2	1	1	1	0	1	2	4	1

Fig. 8. Confusion Matrix on the Test Set

Fig. 8 demonstrates a strong diagonal, with most misclassifications happening between psychologically similar types (e.g.,

INFP–INFJ, ENTP–ENTJ). These confusions, which originate from common linguistic features, are expected and consistent with previous MBTI classification research.

D. Discussion of Results

The 16-class challenge has an accuracy of roughly 86%, proving the potential of voice-based personality recognition. This success is owed to high-quality Whisper transcriptions, a successful hybrid resampling method, contextual power of roberta-base, and a strong training schedule.

The whisper-large-v3 model’s inference latency is the main bottleneck in real-time applications. The method works well for applications with a short delay (e.g., post-interview analysis), but instant-feedback scenarios require tuning. A macro F1-score of 84.72% was achieved, suggesting good and class-balanced performance across all 16 MBTI categories. Whisper processes speech, thus the system remains voice-based. While low-level audio factors (MFCCs, energy, and pitch) were investigated, they produced only minor improvements when compared to gains from ASR transcription and contextual text encoding. As a result, the final design promotes ASR-derived text.

TABLE I
COMPARISON OF PERSONALITY DETECTION MODELS

Model	Description	Accuracy (%)
Mairesse [15]	Linguistic feature extraction + traditional ML	65.2
Golbeck [9]	Twitter-based text classification	69.0
Vásquez [17]	Transformer-based MBTI classification	74.3
Aepli [2]	Cross-lingual transfer with character-level noise	76.5
Akber [3]	Personality + Emotion embeddings	81.0
Bama [4]	Hierarchical Transformer with Label Attention	83.2
Bao [5]	Weighted cross-integrated fusion network	84.5
Our Model	XLNet RoBERTa + Whisper MBTI Classification	85.9

TABLE II
BASELINE MODEL COMPARISON

Model	Accuracy (%)
TF-IDF + SVM	69.7
mBERT	78.3
XLNet-R + Whisper	85.9

V. CONCLUSION

A high-accuracy, voice-based personality recognition system was created by combining OpenAI’s Whisper with a fine-tuned RoBERTa model. The technology accurately classifies MBTI type based on spoken language, providing a modern and non-intrusive alternative to traditional evaluations. This robust performance is the result of a methodology that combines thorough preprocessing with cutting-edge models. This work

lays the groundwork for voice-driven psychological analytics, with potential applications in human resources, education, and digital well-being. Future work will involve expanding multilingual capabilities, including paralinguistic elements, and improving real-time performance.

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